

- If unacceptable side effects persist, discontinue use
- Discontinue concomitant medications that might be contributing to sedation

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- 6–9 g/night in 2 doses, 2.5–4 hours apart

Dosage Forms

- Oral solution 500 mg/mL

How to Dose

- Initial 2.25 g at bedtime and while seated in bed; second dose of 2.25 g should be taken 2.5–4 hours later, also while seated in bed; dosing can be increased by 1.5 g/night every 1–2 weeks; maximum recommended dose 9 g/night



Dosing Tips

- * Both nightly doses of sodium oxybate should be prepared prior to bedtime, and the prepared second dose should be placed in close

proximity to the patient's bed before the first dose is ingested

- Each dose must be diluted with 2 oz of water in a child-resistant cup that is supplied by the pharmacist
- The patient will probably need to set an alarm clock in order to wake up for the second dose
- Once prepared, the solution containing sodium oxybate should be consumed within 24 hours in order to minimize bacterial growth and contamination
- Food significantly reduces the bioavailability of sodium oxybate, so the patient should wait at least 2 hours after eating before taking the first dose of sodium oxybate
- It is best to minimize variability in the timing of dosing in relation to meals
- Once-nightly dosing has been studied and found effective in fibromyalgia

Overdose

- Fatalities have been reported; headache, psychomotor impairment, diaphoresis, vomiting (even when obtunded), confusion, agitation, aggression, ataxia, seizures, coma, death

Long-Term Use

- Long-term use has not been studied
- The need for continued treatment should be reevaluated periodically

Habit Forming

- Medical use of sodium oxybate is classified under Schedule III
- Non-medical use of sodium oxybate is classified under Schedule I
- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses
- History of drug addiction may increase risk of dependence

How to Stop

- Taper may not be necessary when used at therapeutic doses; however, it is possible that some patients could develop withdrawal syndrome

Pharmacokinetics

- Metabolized by the liver
- Elimination half-life approximately 30–60 minutes
- Absorption is delayed and decreased by high-fat meals



Drug Interactions

- Should not be used in combination with CNS depressants or sedative hypnotics



Other Warnings/Precautions

- * Because of the rapid onset of CNS depressant effects, sodium oxybate should only be ingested at bedtime and while in bed
- Sodium oxybate should not be used with alcohol or other CNS depressants, including opiates
- Use only with extreme caution in patients with impaired respiratory function or obstructive sleep apnea
- Patients with history of drug abuse should be monitored closely
- Sodium oxybate may cause CNS effects similar to those caused by other CNS agents (e.g., confusion, psychosis, paranoia, agitation, depression, and suicidality)

Do Not Use

- In patients taking sedative hypnotics
- In patients with succinic semialdehyde dehydrogenase deficiency
- If there is a proven allergy to sodium oxybate (gamma hydroxybutyrate)

Special populations

Renal Impairment

- Dose adjustment is not necessary
- Not excreted renally

- Because sodium oxybate has sodium content, this may need to be considered in patients with renal impairment

Hepatic Impairment

- Reduce dose by one-half and monitor patients closely

Cardiac Impairment

- Not studied
- Because sodium oxybate has sodium content, this may need to be considered in patients with hypertension or heart failure

Elderly

- Safety and efficacy have not been established
- Patients should be monitored closely



Children and Adolescents

- Approved for treatment of cataplexy or excessive daytime sleepiness in pediatric patients 7 years of age and older with narcolepsy



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy

letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women
 - Animal studies do not show adverse effects
 - Placental transfer is rapid; gamma hydroxybutyrate (GHB) has been detected in newborns at delivery following intravenous administration to mothers
 - Use in women of childbearing potential requires weighing potential benefits to the mother against potential risks to the fetus
- * Generally, sodium oxybate should be discontinued prior to anticipated pregnancies

Breast Feeding

- Some drug is present in human breast milk
- * Recommended either to discontinue drug or bottle feed

The art of psychopharmacology

Potential Advantages

- Less activating than stimulants
- For narcolepsy, may help patients insufficiently responsive to stimulants

- Has the biggest effect size on pain, fatigue, and sleep in fibromyalgia

Potential Disadvantages

- Has abuse potential
- Requires a second dose in the middle of the night
- Potentially more dangerous than other treatments, especially for fibromyalgia or chronic pain, and especially if taken with sedative hypnotics and/or opiates

Primary Target Symptoms

- Sleepiness, fatigue
- Cataplexy
- Painful physical symptoms
- Lack of nonrestorative sleep in fibromyalgia and chronic pain

Pearls

- Sodium oxybate increases slow-wave sleep, thereby improving sleep quality at night and allowing individuals to feel more rested during the day
- Sodium oxybate also increases growth hormone, which is one of the reasons that it has been abused by athletes

* There are positive trials for efficacy in fibromyalgia, with improvement in sleep, pain, and fatigue, but an FDA advisory committee voted it down because of safety concerns and the drug is not officially approved for this indication

Suggested Reading

Carter LP , Pardi D , Gorsline J , Griffiths RR . Illicit gamma-hydroxybutyrate (GHB) and pharmaceutical sodium oxybate (Xyrem): differences in characteristics and misuse . *Drug Alcohol Depend* 2009 ;**104** (1–2):1 –10.

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Owen RT . Sodium oxybate: efficacy, safety and tolerability in the treatment of narcolepsy with or without cataplexy . *Drugs Today (Barc)* 2008 ;**44** (3):197–204 .

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solriamfetol

Therapeutics

Brands

- Sunosi
- *see index for additional brand names*

Generic?

No

Class

- Dopamine and norepinephrine reuptake inhibitor

Commonly Prescribed for

(bold for FDA approved)

- **Excessive daytime sleepiness in patients with narcolepsy or obstructive sleep apnea (OSA)**

How the Drug Works

- Boosts neurotransmitters dopamine and norepinephrine/noradrenaline
- Blocks dopamine reuptake pump (dopamine transporter), presumably increasing dopaminergic neurotransmission in brain

regions where the dopamine transporter is present (e.g., striatum)

- Blocks norepinephrine reuptake pump (norepinephrine transporter), presumably increasing norepinephrine neurotransmission in the prefrontal cortex and throughout the brain
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, solriamfetol can increase dopamine neurotransmission in this part of the brain via its actions on the norepinephrine transporter

How Long Until It Works

- Clinical effects can be seen as early as 1 week
- Can take several weeks for some patients to achieve a clinical response

If It Works

- Improves daytime sleepiness
- Treat until improvement stabilizes and then continue treatment indefinitely as long as improvement persists

If It Doesn't Work

Consider an alternative treatment

Could augment with another wake-promoting agent other than stimulants

Best Augmenting Combos for Partial Response or Treatment Resistance

- Best to attempt another monotherapy prior to augmenting with other drugs in the treatment of sleepiness associated with sleep disorders
- If choosing to augment, consider modafinil, armodafinil, or pitolisant

Tests

Before treatment, assess for presence of cardiac disease (history, family history, physical exam)

Assess blood pressure and heart rate at baseline and periodically during treatment

Side effects

How Drug Causes Side Effects

- Side effects are probably caused in part by actions of norepinephrine and dopamine in brain areas with undesired effects (e.g., insomnia, anxiety, headache, increased blood pressure, nausea)
- Side effects are probably also caused in part by actions of norepinephrine in the periphery with undesired effects (e.g., nausea, increased blood pressure, increased heart rate)
- Most side effects are immediate but often go away with time

Notable Side Effects

Headache, insomnia, anxiety

Nausea, decreased appetite

Dose-dependent increase in blood pressure, heart rate



Life-Threatening or Dangerous Side Effects

- Theoretical activation of (hypo)mania, anxiety, hallucinations, or suicidal ideation

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- Lower the dose
- If unacceptable side effects persist, discontinue use

Best Augmenting Agents for Side Effects

Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- Narcolepsy: 75–150 mg once daily upon waking
- OSA: 75–150 mg once daily upon waking

Dosage Forms

- Tablet 75 mg (scored), 150 mg

How to Dose

- Narcolepsy: initial dose 75 mg once daily; can increase every 3 days as needed; maximum dose 150 mg once daily
- OSA: initial dose 37.5 mg once daily; can double dose every 3 days as needed; maximum dose 150 mg once daily



Dosing Tips

Avoid administering within 9 hours of bedtime so that it does not interfere with sleep

In OSA, ensure that the underlying airway obstruction is treated for at least 1 month prior to initiating solriamfetol; modalities to treat the underlying airway obstruction should be continued during treatment with solriamfetol

Overdose

- Limited data

Long-Term Use

- Has been evaluated and found safe and effective in trials up to 1 year
- The need for continued treatment should be reevaluated periodically

Habit Forming

- Schedule IV
- There was no evidence of physiological dependence or withdrawal symptoms in clinical trials

How to Stop

- Taper may be prudent

Pharmacokinetics

- Mean elimination half-life approximately 7.1 hours
- Minimally metabolized; primarily cleared through the kidneys
- Food does not affect absorption



Drug Interactions

Contraindicated with MAO inhibitors, including 14 days after MAOIs are stopped

May affect blood pressure and should be used cautiously with agents that increase blood pressure and/or heart rate as well as with agents used to control blood pressure

Should be used cautiously with dopaminergic agents

Other Warnings/Precautions

Use with caution in patients with bipolar disorder or psychosis

Solriamfetol should be used with caution in patients taking levodopa or amantadine, as these agents can potentially enhance dopamine neurotransmission and be activating

Do Not Use

- If patient is taking an MAO inhibitor
- If there is a proven allergy to solriamfetol

Special populations

Renal Impairment

- Dose adjustment not necessary in patients with mild renal impairment
- Moderate impairment: initial dose 37.5 mg once daily; can increase to 75 mg once daily after 7 days

- Severe impairment: initial and maximum dose 37.5 mg once daily
- Not recommended for use in end-stage renal disease

Hepatic Impairment

- Dose adjustment not necessary

Cardiac Impairment

- Avoid use in patients with unstable cardiovascular disease, serious heart arrhythmias, or other serious heart problems

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women

- When administered to pregnant rats during organogenesis, the no observed adverse effect level (NOAEL) for malformations is 4 times and for maternal and embryofetal toxicity is approximately 1 times the maximum recommended human dose (MRHD) based on mg/m^2 body surface area
- When administered to pregnant rabbits during organogenesis, the NOAEL for malformations and fetal toxicity is approximately 2 times and for maternal toxicity is approximately 5 times the maximum recommended human dose (MRHD) based on mg/m^2 body surface area
- When administered to pregnant rats from gestation day 7 through lactation, maternal toxicity occurred at 7 times the MRHD based on mg/m^2 body surface area; at the maternally toxic dose, fetal toxicity occurred; the NOAEL for maternal and developmental toxicity is approximately 2 times the MRHD based on mg/m^2 body surface area
- Pregnancy registry for solfriamfetol: 1-877-283-6220

Breast Feeding

- Unknown if solriamfetol is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Recommended either to discontinue drug or bottle feed

The art of psychopharmacology

Potential Advantages

- Less abuse potential than stimulants

Potential Disadvantages

- Not available generically

Primary Target Symptoms

- Daytime sleepiness

Pearls

Mechanism of action suggests possible utility in depression, binge eating disorder, ADHD, and smoking cessation, but there are studies only in depression

Monitor blood pressure and heart rate during treatment

Monitor activating side effects (e.g., anxiety, insomnia)

Suggested Reading

Carter LP Carter LP , Henningfield JE Henningfield JE , Wang YG Wang YG , et al. A randomized, double-blind, placebo-controlled, crossover study to evaluate the human abuse liability of solriamfetol, a selective dopamine and norepinephrine reuptake inhibitor . *J Psychopharmacol* 2018 ;**32** (12):1351 –61 .

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Care Med* 2019 ;**199** (11):1421 –31 .
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randomized study of solriamfetol for excessive sleepiness in narcolepsy .
Ann Neurol 2019 ;**85** (3):359 –70 .

Sulpiride

Therapeutics

Brands

- Dolmatil
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine receptor antagonist (D-RAn)
- Conventional antipsychotic (neuroleptic, benzamide, dopamine 2 antagonist)

Commonly Prescribed for

(bold for FDA approved)

- Schizophrenia
- Depression

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis
- Blocks dopamine 3 and 4 receptors, which may contribute to sulpiride's actions
- * Possibly blocks presynaptic dopamine 2 autoreceptors more potently at low doses, which could theoretically contribute to improving negative symptoms of schizophrenia as well as depression

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Continue treatment in schizophrenia until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis in schizophrenia
- For second and subsequent episodes of psychosis in schizophrenia, treatment may need to be indefinite

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics
- Consider trying another conventional antipsychotic
- If 2 or more antipsychotic monotherapies do not work, consider clozapine



Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation of conventional antipsychotics has not been systematically studied
- Addition of a mood-stabilizing anticonvulsant such as valproate, carbamazepine, or lamotrigine may be helpful in both schizophrenia and bipolar mania
- Augmentation with lithium in bipolar mania may be helpful
- Addition of a benzodiazepine, especially short-term for agitation

Tests

- * Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)
- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes

(fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

* Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics

* Monitor weight and BMI during treatment

* While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic

- Monitoring elevated prolactin levels of dubious clinical benefit
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and sulpiride should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
- Antihistaminic actions may cause sedation, weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and any possible increased incidence of diabetes or dyslipidemia with conventional antipsychotics is unknown

Notable Side Effects

- * Drug-induced parkinsonism, akathisia
- * Prolactin elevation, galactorrhea, amenorrhea